

53. Simplify $\left[\sqrt[3]{\sqrt{a^9}}\right]^4 \left[\sqrt[6]{\sqrt[3]{a^9}}\right]^4$; the result is (M.B.A., 2011)

- (a) a^4 (b) a^8
(c) a^{12} (d) a^{16}

54. $(256)^{0.16} \times (256)^{0.09} = ?$ (S.S.C., 2004)

- (a) 4 (b) 16
(c) 64 (d) 256.25

55. $(0.04)^{-1.5} = ?$ (Bank P.O., 2003)

- (a) 25 (b) 125
(c) 250 (d) 625

56. $(17)^{3.5} \times (17)^? = 17^8$ (Bank P.O., 2003)

- (a) 2.29 (b) 2.75
(c) 4.25 (d) 4.5

57. $6^{1.2} \times 36^? \times 30^{2.4} \times 25^{1.3} = 30^5$ (Specialist Officers', 2006)

- (a) 0.1 (b) 0.7
(c) 1.4 (d) 2.6

(e) None of these

58. $2^{3.6} \times 4^{3.6} \times 4^{3.6} \times (32)^{2.3} = (32)^?$ (Specialist Officers', 2007)

- (a) 5.9 (b) 7.7
(c) 9.5 (d) 13.1

(e) None of these

59. $3^{3.5} \times 21^2 \times 42^{2.5} \div 2^{2.5} \times 7^{3.5} = 21^?$ (Bank P.O., 2006)

- (a) 6.5 (b) 8
(c) 10 (d) 12.5

(e) None of these

60. $8^{0.4} \times 4^{1.6} \times 2^{1.6} = ?$ (Agriculture Officers', 2009)

- (a) 48 (b) 52
(c) 64 (d) 76

(e) None of these

61. $8^7 \times 2^6 \div 8^{2.4} = 8^?$ (Bank P.O., 2009)

- (a) 6.6 (b) 8.6
(c) 9.6 (d) 10.6

(e) None of these

62. $25^{2.7} \times 5^{4.2} \div 5^{5.4} = 25^?$ (Bank Recruitment, 2010)

- (a) 1.6 (b) 1.7
(c) 3.2 (d) 3.6

(e) None of these

63. $8^{2.4} \times 2^{3.7} - (16)^{1.3} = 2^?$ (Bank Recruitment, 2010)

- (a) 4.8 (b) 5.7
(c) 5.8 (d) 7.1

(e) None of these

64. $(0.04)^2 \div (0.008) \times (0.2)^6 = (0.2)^?$ (Bank Recruitment, 2010)

- (a) 5 (b) 6
(c) 8 (d) 9

(e) None of these

65. $(18)^{3.5} \div (27)^{3.5} \times 6^{3.5} = 2^?$ (Bank P.O., 2003)

- (a) 3.5 (b) 4.5
(c) 6 (d) 7
(e) None of these

66. $(25)^{7.5} \times (5)^{2.5} \div (125)^{1.5} = 5^?$ (Bank P.O., 2003)

- (a) 8.5 (b) 13
(c) 16 (d) 17.5
(e) None of these

67. The value of $\frac{(243)^{0.13} \times (243)^{0.07}}{(7)^{0.25} \times (49)^{0.075} \times (343)^{0.2}}$ is (C.B.I., 2003)

- (a) $\frac{3}{7}$ (b) $\frac{7}{3}$
(c) $1\frac{3}{7}$ (d) $2\frac{2}{7}$

68. $(64x^3 \div 27a^{-3})^{-\frac{2}{3}} = ?$ (R.R.B., 2006)

- (a) $\frac{9ax}{16}$ (b) $\frac{9}{16ax}$
(c) $\frac{9}{16x^2a^2}$ (d) $\frac{3}{4}x^{-2}a^{-2}$

69. If $2^{n+4} - 2^{n+2} = 3$, then n is equal to

- (a) 0 (b) 2
(c) -1 (d) -2

70. If $2^{n-1} + 2^{n+1} = 320$, then n is equal to

- (a) 6 (b) 8
(c) 5 (d) 7

71. If $3^x - 3^{x-1} = 18$, then the value of x^x is

- (a) 3 (b) 8
(c) 27 (d) 216

72. $\frac{2^{n+4} - 2 \times 2^n}{2 \times 2^{(n+3)}} + 2^{-3}$ is equal to

- (a) 2^{n+1} (b) $\left(\frac{9}{8} - 2^n\right)$
(c) $\left(-2^{n+1} + \frac{1}{8}\right)$ (d) 1

73. The value of $\frac{2^{3x+4} + 8^{x+1}}{8^{x+1} - 2^{3x+2}}$ is

- (a) 3 (b) 4
(c) 5 (d) 6

74. The value of $\frac{2^{n-1} - 2^n}{2^{n+4} + 2^{n+1}}$ is

- (a) $-\frac{1}{36}$ (b) $\frac{2}{3}$
(c) $\frac{1}{13}$ (d) $\frac{5}{13}$